



An Improved Harmony Search Approach for Block Placement for VLSI Design Automation

Shaik Karimullah¹ · D. Vishnuvardhan² · Vidhyacharan Bhaskar³ 

Accepted: 8 June 2022 / Published online: 1 September 2022

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2022

Abstract

Level of knowledge with the field for VLSI IC Design's Placement and Routing and in the area of Wireless Communication is rapidly evolving; Hence the process designing in the above fields is critical to assimilate a higher quantity of computation elements or nodes into a very compact size in VLSI Area and also the same is applicable in Wireless Communications to cover more number of nodes in a specific area. Prior to completing the placement, the physical and technical arranging of the computation elements or nodes in the chip area is planned, which is nothing more than Placement step and analogous with coverage of nodes in Wireless communication. Floorplanning must be done properly for reduce given placement area in physical ICDie. The proposed work suggests an ImprovedHarmonySearch-IHS Optimization Algorithm, which is an Sequential combination of the HarmonySearch (HS) Algorithm and the most effective Particle Swarm Optimization (PSO) Algorithm to achieve the major objective of the VLSI Placement approach and applicable to the field of Wireless Communication. The aim at this point is to reduce the gap between the standard area of the benchmark circuit to the obtained area of the corresponding benchmark circuit. The MATLAB code for the ImprovedHarmonySearch-IHS Algorithm is recreated, and discoveries are tested for better inspection using standard MCNC, or North Carolina Microelectronics Center benchmark circuits and suggest to adopt the corresponding results to the area of Wireless communication.

Keywords Wireless Communication · Optimization · Physical Design · ImprovedHarmonySearch-HIS

1 Introduction

As market mandate for innovative expertise grows, the IC manufacturing division has performed additional research, resulting in VLSI enterprise that is more multiple, more dependable, additional close, besides produces improved outcomes. Because there are so many transistors in one chip, the IC structural plan has become ample extra intricate. Placement planning is an additional intricate method to the aforementioned problem, and also a key

Extended author information available on the last page of the article

design segment for hierarchical, building block design technique. As the aspect ratio of the circuit grows, so does the size of the perseverance, making the global elucidation difficult to interpret [1]. Following apportioning, the multifarious circuit is divided and hooked on minor circuits in the next step, known as floor layout.

It is recycled to compute the comparative placements of the numerous building blocks in every sub-circuit in command to compress the overall vacant area for any proposal of the given circuit besides for the decrease the overall space in chipdie occupied by design. Apportioning, Placement, locating and routing are all processes in the physical design of IntegratedCircuits. The VLSI strategy procedure is critical in the production of integrated circuits. Placement is an important stage in the computerization of electronic circuit plan; it is an element of the physical plan movement that allocates precise positions inside the chip's primary region is used to house numerous circuit parts [2], [3]. This may result in a lower-level arranging profession being unproduced if the wire length exceeds the routing capabilities available to the ICchip. For considered system design to placate its performance standards, a placer is needed to finish the work and optimize a variety of goals. Mathematical processes (such as a function maximum) are employed in this context to create something (such as a design) that is both functional and efficient [4], [5]. Applying mathematical concepts and techniques to solve quantitative issues in fields such as physics, biology, engineering and economics is the focus of applied mathematics. The unchanged evaluation tools can be used to tackle problems involving exploiting or underrating characteristics for a variety of domains.

HarmonicSearch stands a meta-heuristic procedure for music improvement [6]. The view was prompted by the assumption that music is seeking an ideal condition. This settlement in music remains similar to optimality popular a improvement sequence. The expedition method aimed at improvement can be compared to the unplanned production process of a jazz musician [7][8][9]. The standard of sound style, from one point of view, indicates the perfect accord. A performer must always provide completely pleasing music. Both approaches are intended to provide the greatest or best possible results. Such similarities between two techniques can be exploited to generate new algorithms by learning from one another [10], [11]. The best example of Through ingenuity, the beauty and consistency of music is transformed hooked on a routine of improvement with identifying a unadulterated harmony, which is called HarmonySearch [12], [13].

2 Associated Work

New plan patterns need emerged as VLSI technology has advanced. Because to element dimension and symbol concede reductions, completely available time postponement currently associates to connectivity interruptions during 80% of the complete ICchip capacity. Through this detailed situation, planning is pleasant increasingly important, besides innovative solutions are compulsory [14, 15]. The basic aspect of the spectrum fluctuating tactics are that they offer a variety of alternatives for identical cataloguing. These stand a tremendous help in defeating adjacent optima [16, 17]. Particle Swarm Optimization [18, 19] is one solution for VLSI position. When the underlying circumstance is ideal, this methodology is very practical.

Mr. Maji and colleagues [20] proposed a research calculation called as Crazy dependent PSO aimed at simplifying the Placement of blocks in VeryLargeScaleIntegration of chips. It remained used towards expedite local queries too increase the accuracy of the prearrangement. The primary purpose of placement arrangement development is towards reducing the IC area plus entire connecting path. Laudis and colleagues [21] focused on the complex mark bat design (MOBA), a certainly triggered contraction, and successfully exploited that one to restore the Placement arrangement in VLSI IC design. The scheming's main idea is toward advance the situation by mimicking bats' echolocation abilities. The problem was viewed as a MOO-MultiObjectiveOptimization problem, with wirelength reduction and dead-space deduction given equal weight.

Basir and Kazeruni [22] offered additional temperature susceptibility particularizing throughout the clocking tree incorporation quantity called Stochastic perturbation based clock reorganization to advance the nature of a VLSI execution (SPECO). The investigation looks on the effect of a ICchip's heat abnormality on synchronisation excellence, that impacts the speed of modern VLSI devices. The fact-finding results revealed a significant reduction in dephasing and computational convolution. Kazeruni and Basirand developed alternative heat defencelessness strategy through the clockingtree grouping method termed stochastic annoyance-based clock enhancement to increase the nature of a VLSI usage (SPECO). The investigation takes into account the effect of a ICchip's hotness difference on synchronisation superiority, that impacts the rapidity of modern VLSIIC processors. The investigative results revealed a significant reduction in the dephasing and computational multidimensional character.

Following that, Yang and his colleagues [23] predicted another blended method that combined the insect state calculation with the banned inquiry calculation to improve the net directing plan conspiracy in VLSI actual plan. The results showed that the novel method avoided the small permutation proportion in the fundamental time of an important below ground inspect domain background. When compared to available techniques, the competence of the banned clandestine inspects domain structure enhanced by approximately 17.567%. Furthermore, it successfully avoided adjacent optima. Mr. Subhojit and colleagues [24], [25] presented an expert technique for designing a fixed-request compensator for compensating the current-mode control design of DC-DC converters. The proposed computation combined a population exploration centred development technique, obviously, PSO, with a neighbouring analysis centred strategy. Despite the fact that he and his colleagues have sought to ameliorate the VLSI scenario by devising new computations and techniques, with maximum evaluation complexity concern have not been addressed in recent times. Martins and colleagues [26] focused with scheduled the concept of MOO although automating the prearrangement in a basic synchronized circuit structure proposal. To improve the position of each nearness gathering, an imaginative organizer centred multiobjective imitated assuaging intention functioning above an entire description is presented. Instead of the traditional solitary objective arrangement comes adjoining, the successive Pareto frontages conciliations amongst the streamlining purposes of every single congregation are associated, dishonourable up, concluded the strategy advanced method, awaiting a final facade is obtained. As a result, the difficulty's unpredictability is reduced and distributed among multiple performances of the development phase through fewer design elements.

Taher, et al. [26] proposed PASSIOT, a new attitude for various independent and upgraded synthesis of basic designs based on imagining lists aimed at their changeable

boundary trajectories. PASSIOT remained reacted using unpretentious restructuring abilities and stayed to shown to be significant in terms of runtime and arrangement quality. An adaptable mixture memetic scheme was familiarised in [27] to exaltation the area, entire switching path, and utmost exciting and usual heats of a ICchip in order to enhance productively in a multiple unprejudiced thermally aware non-slicing Placement assembling approach. The traditional examination scheming was used for global expedition strategy in the proposed calculation of [28] to study the chase space as much as possible. As a nearby hunt strategy, an altered reproduced strengthening search calculation is used to misinterpret the facts in the inquiry [29] [30]. A passing likelihood system adjusts the global investigation and local manipulation. The explained method successfully obtained Placement designs thus lowering the usual and maximum hotness, according to trial findings on common test benchmarks.

3 Objective of Algorithm

Positioning a block will be an important consideration in mechanization of Very Large Scale Integrated circuits (VLSI) and architecture since this affects the efficacy, volume, also performance of VLSI IC circuits, and their trustworthiness. VLSI Positioning is an issue in terms of computing. The existing integrating technique with extremely large perspectives is related with fixed mode placement limits along with constrained region and wire length interconnecting the modules. The utilization of every succession square design modules up to lower identified and highlighted in connection with maximum interface wire length are the major changes in the permanent form of VLSI circuits. Many of the modules are still in the early stages of development and are frequently adjustable when it comes to moulding circuit elements (or mega cells). A circuit component made up of twelve blocks are arranged with a line twelve modules, two rows with five components, three rows of eight components, and so on, giving the layout design a number of implausible configurations in that architecture.

Assume a rectangular locale S with size Y and prestige X , a huge number of components $N=n_1, n_2, \dots, n_n$, wherein component n_i is a quadrangular corner with a specified size Y_a size X_i as well as a gross quick overview M_i indicating interlinkages between components in N , the problem for explore a placing of all components in the quadrangular region S so those who achieve certain expectations:

1. Components not to be linked horizontally; they must be linked perpendicular to the coordinate's axis.
2. It is not allowed to combine one component with another.
3. Components must be setup in S .
4. All components can be turned 180 degrees while placement.

4 Proposed Algorithm

4.1 ImprovisedHarmonySearch-IHSAlgorithm

The effective method is to attempt and test the approach for mixing between them in order to improve the display and viability of the analyses. The ImprovisedHarmonySearch-IHS is the product of a crossbreeding of the HarmonySearch(HS)Algorithm & ParticleSwarmOptimization. The mixing is completed by implementation with a dropping approach for both the Harmony Search (HS) Algorithm and the Particle Swarm Optimization (PSO), with the dropping done advancing. Because the individual performances of HS and PSO are so good, they are merged. HS is a rapid method that produces consequences rapidly, Though PSO has a slower conjunction rate hence it produces nearly grander outcomes. IHS Algorithm is implemented with the aim for Designing in VLSIICPhysicalSystems Engineering for preserve both precision and agility. Harmony Search is a way of optimizing by identifying a perfect harmony by translating qualitative experimentation into quantitative rules through innovation, and therefore the elegance and consistency of music into a technique of optimization.

4.2 Sequential steps of IHS Algorithm:

ImprovisedHarmonySearch (IHS) Algorithm's Fig. 1 depicts the operational flow. The procedure consists of two stages: execution and simulation. In the first stage, the fundamental algorithms, a HarmonySearch(HS) Technique, would have developed with tested inputs. When the stopping requirement is fulfilled, the HSAlgorithm do run as per the sequence of steps specified, such as HMCR_HarmonyMemoryConsideringRate, PAR_PitchAdjustingRate, and Bandwidth (BW), for providing the best and optimum solution afterward the necessary numeral of iterations. Here first stage procedure then will be completed and carried out. The second phase of operations will be invoked when the first step has been computed. The first-stage solutions should be used as contributions in the second step. To outperform the standalone algorithms, the normal PSO approach will be followed in the phase two of said Improvised Harmony Search (IHS) Algorithm. To obtain the best results, HS as well as PSO will be run one by one in a sequence configuration. The proposed method is projected to overcome numerous disadvantages of the suggested algorithms based on exactitude and required presentation.

4.3 Algorithm stages in the ImprovisedHarmonySearch (IHS) Approach:

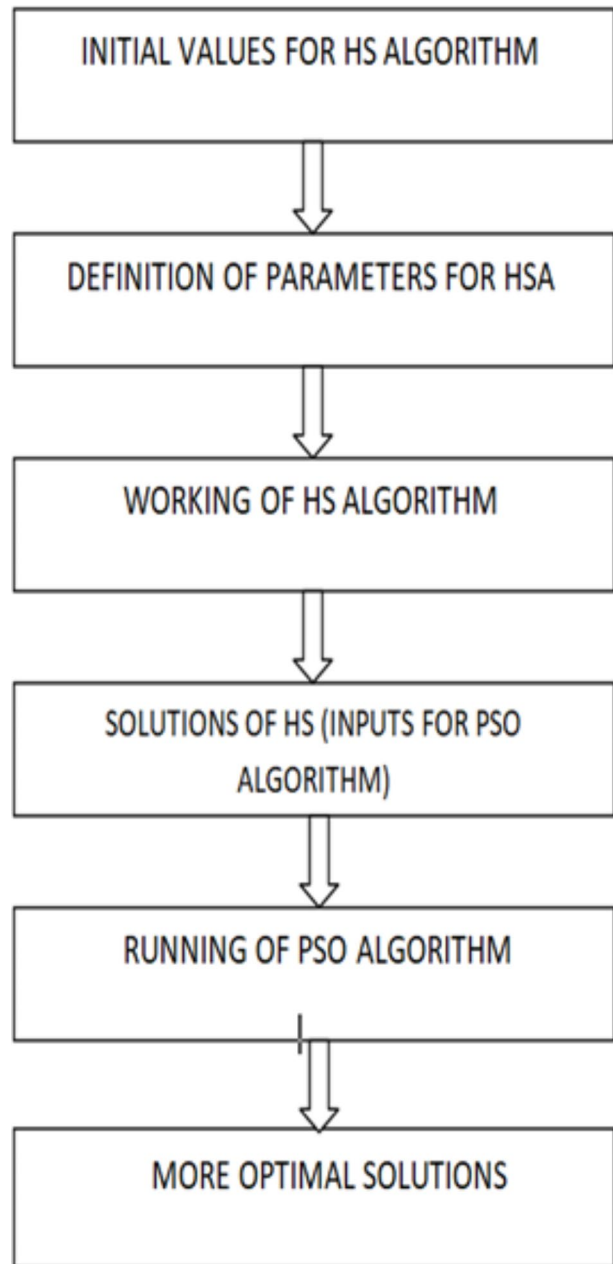
The following is a step-by-step approach for the I H Search (IHS) Algorithm:

1. Initialize the harmony's memory.
2. Configure the HMCR, PAR, and BW settings.
3. Run the process for finding optimum result.
4. The aforesaid IHS answers are used as beginning values by the P S O approach.

PSO will develop and implement more optimal solutions.

The process will continue until the optimal solution is discovered.

Fig. 1 Sequential steps of IHSAlgorithm



4.4 Harmony Search (HS) Approach:

Geem, et al. proposed the Harmony Search (HS) meta-heuristic optimization technique. It is modelled after the practice of musical experimentation; in which musicians employ musical improvisations to strive to achieve harmonious relationship in an orchestra or band. When

musicians create harmonics, the sound pairings recorded with the memory typically attempt for change. This step by step programme created for replicate how well the performer uses recollection and furthermore previous experience to arrive at the note that results in the most satisfying harmony after living beside the opposites. HS shows simple steps to execute and used to correct virtually any hindrance that could arise as a result of a reduced or increased purpose. The strategy of obtaining optimum or relatively close resolutions with an entanglement will employed to bearing every mathematical inquiry of an ideal condition of coherence. The additional solution is only momentary because it is based on three rules.

1. The first consideration is memory.
2. Pitch adjustment.
3. Choosing at random.

The following are the HS Algorithm's major control parameters:

1. Harmony Memory (HM).
2. Memory Size Harmony (HMS).
3. Harmony Cognition Taking Rate (HMCR).
4. RateofPitchAdjustment(PAR).
5. BW_Bandwidth.

4.5 Process Flow for ImprovisedHarmonySearch-IHSApproach:

Process Flow of ImprovisedHarmonySearch-IHSApproach is shown below:

Figure 2 is a standard ImprovisedHarmonySearch-IHS flowchart, demonstrating how the method begins and ends. The region of use is used to generate the underlying congruity memory. Concurrently, the concordance search bounds are established, as well as the Pitch Adjusting Rate (PAR), Evaluation Rate using Harmony Memory. The asymmetrical value is selected as a prominent parameter of a certain loop. If astronomical number is below the HMCR, therefore randomly chose a present symphony if indeed the odd amount is more than the PAR, then randomly alter the pitch within cutoff points; randomization would provide new music in either scenario. The process will be continued till the stopping measurement is achieved, i.e. till the group is completely in sync.

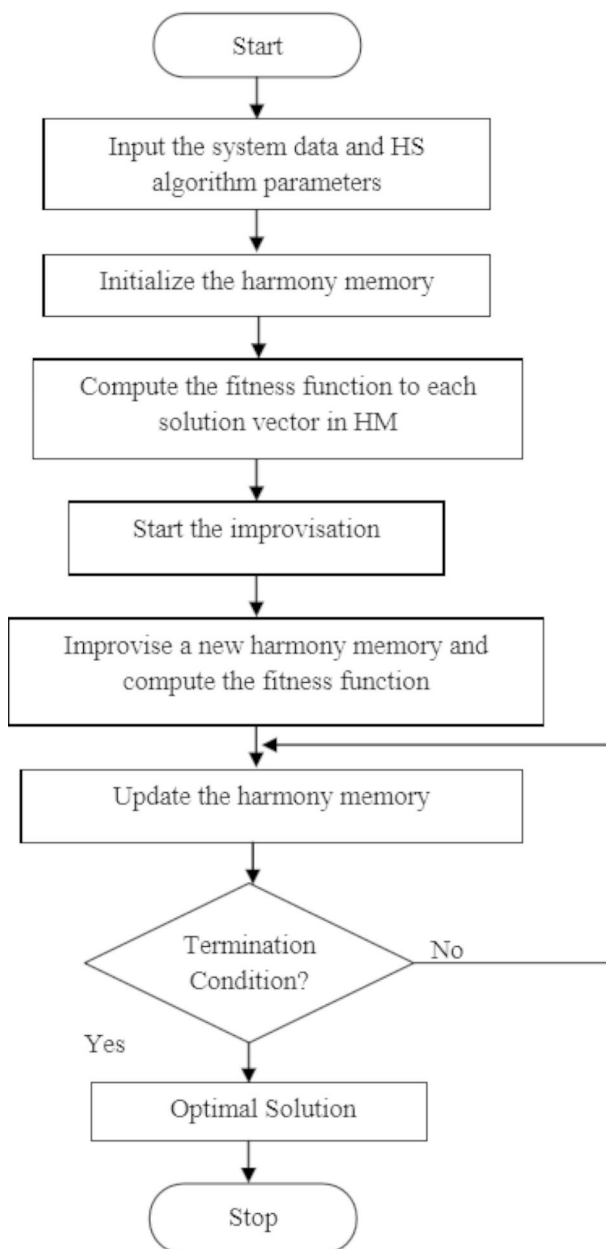
4.6 Steps in Harmony Search Algorithm

Step1: Update value of the HS(HM). The original HS(HM) is made up with random set of ratings allocated towards optimization current discourse.

Step2: Enhance HM's novel layout. Each element, this arrangement has been acquired, according to the HMCR. The HMCR is defined as the probability that a section will be picked at random first from present HM population, and thus 1HMCR is probability with gaining by chance.

Step3: You must upgrade the HM(HS). An Updated Phase two architecture now being evaluated. If it offers preferable health over the most notably awful piece, it replace memory value. If not, this discarded.

Fig. 2 Process Flow of Improvised-HarmonySearch-IHSApproach



Sep4: Repetition of Stages 3 and 2 until a predefined termination condition is satisfied. Parameter Definition for the Harmony Search Algorithm.

The HS Algorithm makes extensive use of Harmony Memory, and the greatest harmonies are moved Harmony Memory. To make better use of this memory, typically given as adjustable named HarmonyMemoryConsideringRate_(HMCR). If indeed the HMCR is small, only a several of the finest harmonies will be identified, and the uniting process may be

Table 1 Values of typical MCNCBM Architectures

BMArchitecture	Typicalarea(mm ²)	cells	nets
AMI49	35.4	49	408
HP	08.8	11	83
XEROX	19.33	10	203
AMI33	1.1	33	123
APTE	46.5	9	97

excessively slow. If the HMCR's is extremely great, essentially all of coherences are used for memory of a congruency, resulting in certain suboptimal arrangements since different harmonies are not thoroughly investigated. HMCR normally varies between 0.7 and 0.95. HMCR could be set to 0.9 in this study to increase the method's performance throughout Placement. Pitch modification is governed by BW_spectrum (BW) and pitch adjustment rate (PAR). The amount of change is controlled by PAR, which normally varies between 0.1 and 0.7. The benchmark is 0.3 and also the SpectrumBW is 0.1 for a smooth operation.

5 Discussion and results

Simulating the code with the proposed technique with mention to the reference value of MCNCBM Architectures yields the optimum result for cell/block placement. The simulation results carried out on basic MCNC Benchmark architectures to show placement of blocks clearly, however this algorithm is capable of dealing with architectures having hundreds blocks like any processor architectures. The simulation results for the processor architectures leads to more confusion to analyze and relatively existing algorithms also applied on MCNC architectures to evaluate their efficiency. So this work is carried out on the basic and less block architectures and IHS Algorithm shows proportionate change on any architecture with any number of blocks. The values of the reference characteristics of conventional MCNCBM circuits are shown in Table 1.

The results of ImprovisedHarmonySearch-IHS Algorithm to attain improved augmented Placement outcomes are shown beneath:

The simulation outcome for the Benchmark architecture the Apte with 9 blocks set using ImprovisedHarmonySearch-IHS Algorithm is shown in Fig. 3. This yields results for an ideal area of 46.5mm² and a simulated area of 47.5mm² using ImprovisedHarmonySearch-IHS. The placement of 9 blocks using the IHS Algorithm with block number 9 after 4 yields the optimal placement with the least amount of area occupied. As this architecture have symmetrical shaped blocks with only one small block, placement made possible by using IHS Algorithm in easy way. As shown in Fig. 3 block 9 have been placed between block 4 and block 6 to achieve optimum result.

The simulation outcome for the Test Xerox with 10 cells added using the.

ImprovisedHarmonySearch-IHS Algorithm is shown as Fig. 4. This value yielded a typical area with 18.355mm², which was executed with ImprovisedHarmonySearch-IHS to yield a total area of 20.585 mm².

The simulated outcome for the Benchmark architecture the HP with 11 blocks arranged using Improvised Harmony Search-IHS Algorithm is shown in Fig. 5. This value yielded a potential point of 8.983mm², which was simulated with an IHS of 10.495 mm².

Fig. 3 Simulation Results of MCNCBMCAPTE

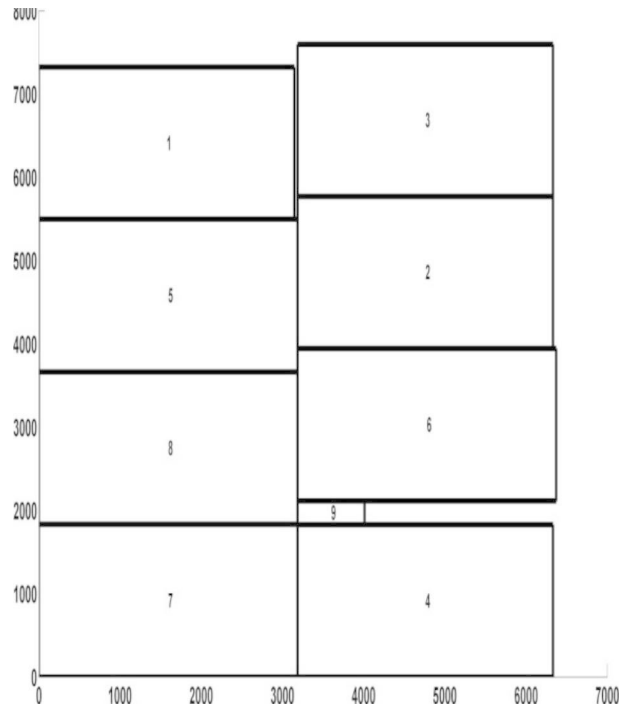
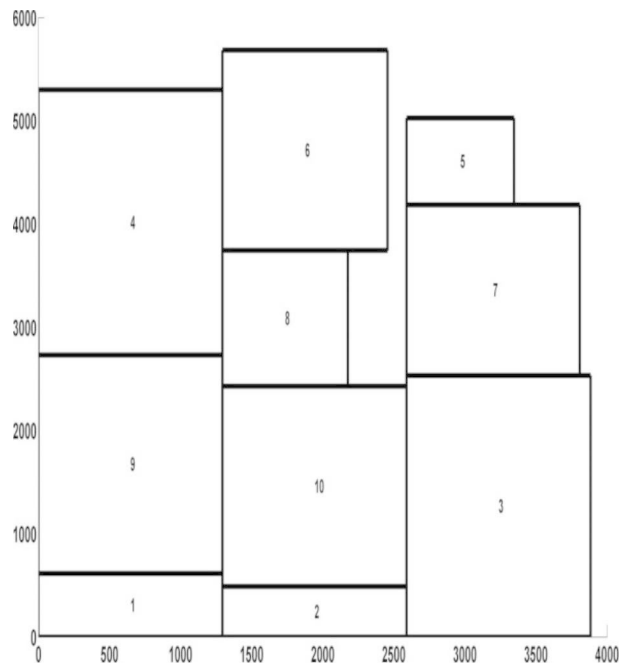
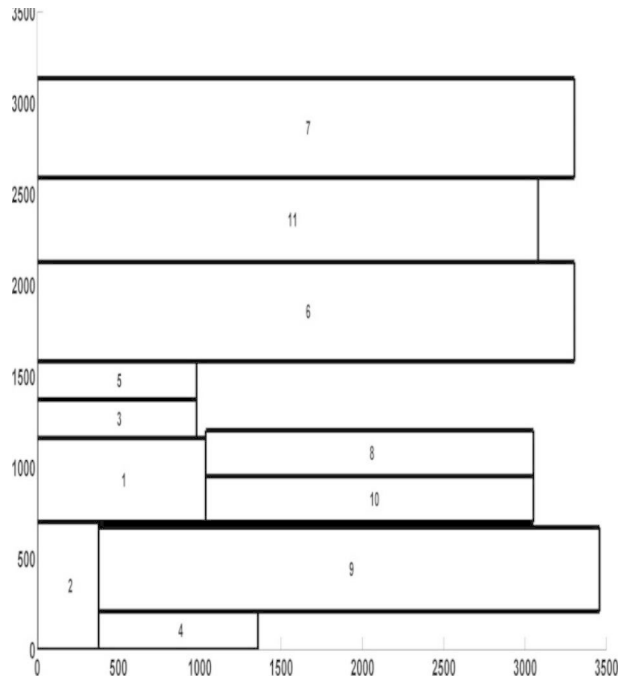
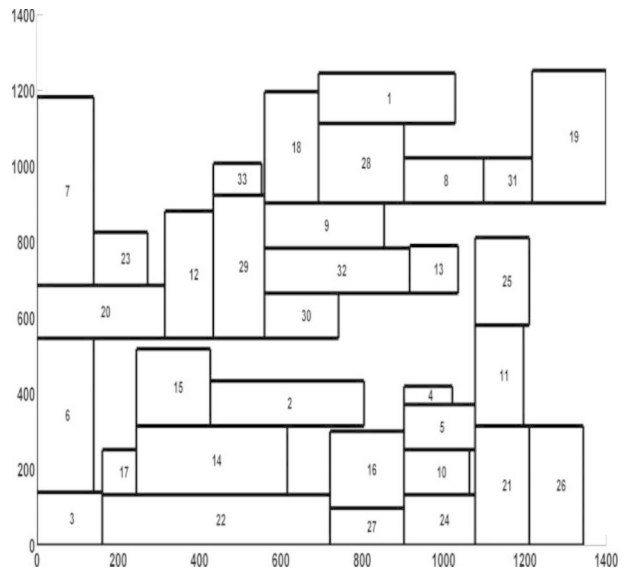


Fig. 4 Simulation Results for MCNCBMC XEROX

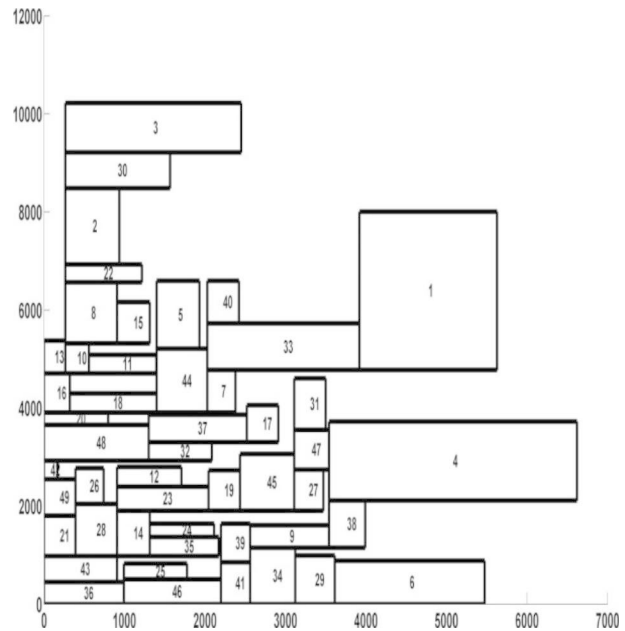


Using the ImprovedHarmonySearch-IHS Algorithm, the Benchmark AMI33 with 33

Fig. 5 Simulation Results of MCNCBMC HP**Fig. 6** Simulation Results of MCNCBMC AMI33

cells is presented in Fig. 6. An optimum area of 1.115 mm² has been simulated using an IHS of 1.0366 mm² with this value.

Simulated results for AMI49 with 49 cells arranged using the ImprovisedHarmonySearch-IHS Algorithm are shown in Fig. 7. A simulated area of 61.932 mm² was created using an IHS of 35.144 mm² as the optimal area.

Fig. 7 Simulation Results of MCNCBMC AMI49**Table 2** Comparison of HS with IHSAlgorithm

Benchmark circuit.	Number of cells	Ideal area in mm^2	Simulated area with HS in mm^2	Simulated area with IHS in mm^2
Ami49	49	35.144	65.5	61.932
Ami33	33	1.115	1.77	1.366
Apte	9	45.56	48.21	47.505
Xerox	10	18.355	21.885	20.585
Hp	11	8.983	10.54	10.495

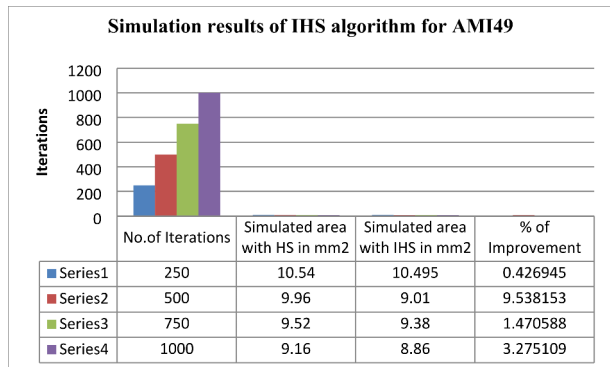
Predefined MCNCB circuits are shown in the results above, with modules arranged among length and even the breadth in vertical ways. As shown in Fig. 3, the Placement of nine components (APTE) is shown in the defined Xhub courses and Yhub department, and it was obtained after successively the dispersed IHS Algorithm code with every possible amount of emphasises. To illustrate this point, the Placement of 33 units, 49 units, and ten modules are shown in Figs. 4, 5 and 6, and 7.

Table 2 shows the examination investigation of area acquired by the use of IHS Algorithm and the HS Algorithm alongside the standard region of MCNCB Circuits. It tends to be seen from the Table 2 that the dead space rate is similarly low and the territory got is exceptionally fulfilling if there should be an occurrence of proposed calculation which the sign of a decent Placement.

After successful simulation of IHS algorithm for various benchmark Circuits, The algorithm simulated for 250,500,750,1000 iterations respectively and compared its performance with HS algorithm, corresponding values are as shown below.

Table 3 Comparative Simulation results of IHS algorithm for AMI49

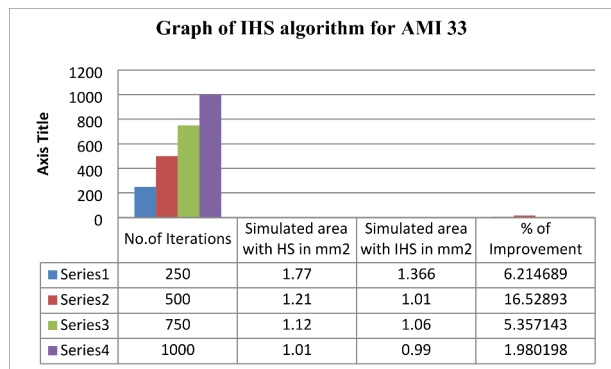
Benchmark circuit.	Number of cells	Ideal area in mm ²	No. of Iterations	Simulated area with HS in mm ²	Simulated area with IHS in mm ²	% of Improvement
Ami49	49	35.144	250	65.5	61.932	5.447328
			500	64.98	63.1	2.893198
			750	64	63.12	1.375
			1000	63.82	63	1.284864

Fig. 8 Simulation graph of IHS algorithm for AMI49**Table 4** Comparative Simulation results of IHS algorithm for AMI 33

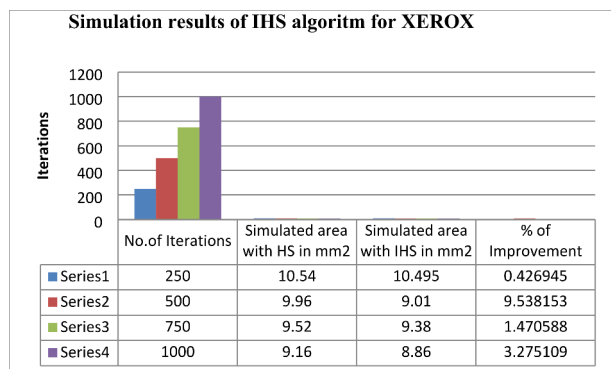
Benchmark circuit.	Number of cells	Ideal area in mm ²	No. of Iterations	Simulated area with HS in mm ²	Simulated area with IHS in mm ²	% of Improvement
Ami33	33	1.115	250	1.77	1.366	6.214689
			500	1.21	1.01	16.52893
			750	1.12	1.06	5.357143
			1000	1.01	0.99	1.980198

By simulating the IHS algorithm With for AMI 33 circuit with 33 blocks achieved 6.21% of improvement for 250 iterations, 16.5% for 500 iterations, 5.35% for 750 iterations, 1.98% for 1000 iterations in comparison with HS algorithm and the corresponding relationship also shown with the help of below graph, Fig. 9. Using the proposed strategy, the placement of blocks is significantly improved for every 250 iterations and reaches its optimum

By simulating the IHS algorithm With for AMI 49 circuit with 49 blocks achieved 5.4% of improvement for 250 iterations, 2.8% for 500 iterations, 1.37% for 750 iterations, 1.28% for 1000 iterations in comparison with HS algorithm and the corresponding relationship also shown with the help of below graph, Fig. 8. For every 250 iterations the proposed method shows significant improvement for placement of blocks and achieved optimum result. With 250 iterations we can get 5.4% of improvement in placement of blocks, from 250 to 500 iterations the algorithm attained 2.8% of improvement, whereas from 500 to 750 and from 750 to 1000 iterations the proposed IHSAlgorithm achieved nearly 1.3% of improvement as shown in Table 3 and Table 4. As demonstrated in Table 3, the suggested IHSAlgorithm improved the place-

Fig. 9 Simulation graph of IHS algorithm for AMI33**Table 5** Comparative Simulation results of IHS algorithm for XEROX

Benchmark circuit.	Number of cells	Ideal area in mm ²	No. of Iterations	Simulated area with HS in mm ²	Simulated area with IHS in mm ²	% of Improvement
Xerox	10	18.355	250	21.885	20.585	5.940142
			500	20.16	19.26	4.464286
			750	19.55	19	2.813299
			1000	19.21	18.95	1.353462

Fig. 10 Simulation graph of IHS algorithm for XEROX

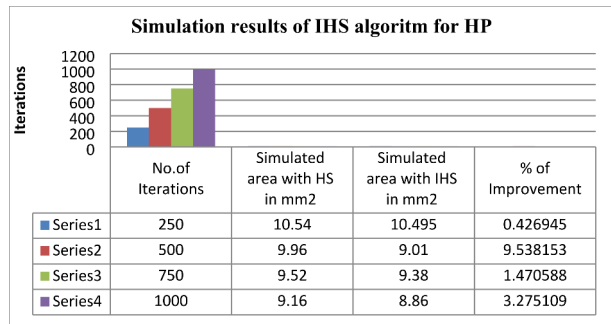
ment of blocks by 6.2 percent after 250 iterations, by 16.5percent after 500 iterations, and by 5.3 percent after 750 iterations and very less at 1000 iterations (Table 5).

By simulating the IHS algorithm With for Xerox circuit with 10 blocks achieved 5.94% of improvement for 250 iterations, 4.46% for 500 iterations, 2.81% for 750 iterations, 1.35% for 1000 iterations in comparison with HS algorithm and the corresponding relationship also shown with the help of below graph, Fig. 10.

Every 250 cycles, the suggested technique exhibits a considerable improvement in block placement and achieves the best outcome. With 250 iterations, we can get 0.42% increase

Table 6 Comparative Simulation results of IHS algorithm for HP

Benchmark circuit.	Number of cells	Ideal area in mm ²	No.of Iterations	Simulated area with HS in mm ²	Simulated area with IHS in mm ²	% of Improvement
Hp	11	8.983	250	10.54	10.495	0.426945
			500	9.96	9.01	9.538153
			750	9.52	9.38	1.470588
			1000	9.16	8.86	3.275109

Fig. 11 Simulation graph of IHS algorithm for HP

in block placement, 9.5% improvement from 250 to 500 iterations, and approximately 1.4, 3.2% improvement from 500 to 750 and 750 to 100 iterations respectively.

By simulating the IHS algorithm with for Hp circuit with 11 blocks achieved 0.42% of improvement for 250 iterations, 9.5% for 500 iterations, 1.47% for 750 iterations, 3.27% for 1000 iterations in comparison with HS algorithm and the corresponding relationship also shown with the help of below graph, Fig. 11.

After 250 cycles, the suggested approach significantly improves block placement and achieves the optimal outcome. With 250 iterations, we can achieve a 0.42% improvement in block placement, 9.5% improvement with 500 to 500 iterations, and approximately 1.4, 3.2% improvement with 750 to 100 iterations respectively, as shown in Table 6.

6 Conclusion and Future Enhancements

An ImprovisedHarmonySearch-IHS is used in this research to estimate how much space can be saved in Benchmark designs by using different iterations, and this method is also used for the notion of node location in Wireless Communications at the same time. We can conclude from this calculation that the suggested method is more efficient at evaluating area than some other continuous improvement such as single PSO and single HS and that the range of stages and levels it establishes is likewise less numerous. A new domain of MCNCBM Circuits has now been achieved by utilising this method, and we may compared it to the usual zone. Analysis shows that the proposed computation is able to meet the objectives and goals of Optimizing Placement, thereby providing a better basis for VLSI Design. " This proposed computation reduces the amount of dead space in the VLSI layout, making it more important for various types of testing in the area.

Authors' contributions **Shaik Karimullah:** Conceptualization, Methodology, Writing - original draft, Analysis. **D.Vishnuvardhan:** Visualization, Investigation, Validation, Editing, Supervision. **Vidhyacharan Bhaskar:** Supervision, Writing - review & editing.

Funding None.

Data Availability The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to containing information that could compromise the privacy of research participants.

Code Availability The code that support the findings of this study are available on request from the corresponding author. The codes are not publicly available due to containing information that could compromise the privacy of research participants.

Declarations

Conflicts of interest/Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Chandrasekaran, G., Periyasamy, S., & Rajamanickam, K. P. (2019). Minimization of test time in system on chip using artificial intelligence-based test scheduling techniques. *Neural Computing and Applications*, 1–10
- Wu, C., Fang, J., & Li, Q. (2019). Multi-material topology optimization for thermal buckling criteria. *Computer Methods in Applied Mechanics and Engineering*, 346, 1136–1155
- Sivasubramanian, K., & Jayanthi, K. B. (2015). Music-inspired harmony search algorithm for fixed outline non-slicing VLSI floorplanning. *World Academy of Science, Engineering and Technology, International Journal of Electrical, Computer, Energetic, Electronic and Communication Engineering*, 9(6), 512–519
- Gwee, B. H., & Lim, M. H. (1999). A GA with heuristic-based decoder for IC floorplanning. *Integration*, 28(2), 157–172
- Shaik Karimullah, & D. Vishnu Vardhan (2022). Pin density technique for congestion estimation and reduction of optimized design during placement and routing. *Applied Nanoscience*.
- Fernando, P. R. (2009). Genetic algorithm based design and optimization of VLSI ASICs and reconfigurable hardware
- Grefenstette, J., Gopal, R., Rosmaita, B., & Van Gucht, D. (1985, July). Genetic algorithms for the traveling salesman problem. In *Proceedings of the first International Conference on Genetic Algorithms and their Applications* (Vol. 160, No. 168, pp. 160–168). Lawrence Erlbaum
- Karimullah, S., & Vishnuvardhan, D. (2018) A review paper on optimization of placement and routing techniques. *International Journal of Engineering and Techniques*.
- Rebaudengo, M., & Reorda, M. S. (1996). GALLO: A genetic algorithm for floorplan area optimization. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 15(8), 943–951
- Venkatraman, S., & Sundhararajan, M. (2017). Optimization of VLSI floorplanning using genetic algorithm. *J Chem Pharm Sci JCPs*, 10(1), 311–316
- Karimullah, S., Vardhan, D. V., & Basha, S. J. (2020). Floorplanning for Placement of Modules in VLSI Physical Design Using Harmony Search Technique. *ICDSMLA 2019* (pp. 1929–1936). Singapore: Springer
- Yang, X. S. (2010). Biology-derived algorithms in engineering optimization. arXiv preprint arXiv:1003.1888
- Funke, J., Hougardy, S., & Schneider, J. (2016). An exact algorithm for wirelength optimal placements in VLSI design. *Integration*, 52, 355–366
- Kureychik, V., & Kulakov, A. (2017, December). Algorithm of Thermal Optimization of Placement of Basic Elements of VLSI. In *IV International research conference "Information technologies in Science, Management, Social sphere and Medicine (ITSMSSM 2017)*. Atlantis Press
- Chandrasekaran, G., Periyasamy, S., & Karthikeyan, P. R. (2019). Test scheduling for system on chip using modified firefly and modified ABC algorithms. *SN Applied Sciences*, 1(9), 1079

16. Shaik Karimullah, S. J., Basha, P., Guruvyshnavi, K., & Sathish Kumar Reddy (2020). *B Navyatha "A Genetic Algorithm with Fixed Open Approach for Placements and Routings"* ICCCE (pp. 599–610). Publisher Springer
17. Vishnuvardhan, D. D., Shaik Karimullah" Monitoring Physiological Variables Of Mining Workers" Ncetimes#1.0, Ijet Issn:2395-1303-2018
18. Jeevitheswari, B., S. Karimullah, Vishnu Vardhan (2017). A Low Power DFT Based CMOS Receiver with Dual Use of Power Lines" *IJSETR* 6(22), 4470–4475
19. venkatraman, S., & Sundhararajan, M. (2017). Particle Swarm Optimization Algorithm For VLSI Floorplanning Problem. *J Chem Pharm Sci Jcps*, 10(1), 311–316
20. Xue, B., Zhang, M., & Browne, W. N. (2012). Particle swarm optimization for feature selection in classification: A multi-objective approach. *IEEE transactions on cybernetics*, 43(6), 1656–1671
21. Maji, K. B., Ghosh, A., Kar, R., Mandal, D., & Ghoshal, S. P. (2015, November). An evolutionary algorithm based approach for VLSI floor-planning. In 2015 International Conference on Science and Technology (TICST) (pp. 248–253). IEEE
22. Laudis, L. L., Shyam, S., Jemila, C., & Suresh, V. (2018). MOBA: multi objective bat algorithm for combinatorial optimization in VLSI. *Procedia Computer Science*, 125, 840–846
23. Basir-Kazeruni, S., Yu, H., Gong, F., Hu, Y., Liu, C., & He, L. (2013). SPECO: Stochastic Perturbation based Clock tree Optimization considering temperature uncertainty. *Integration*, 46(1), 22–32
24. Yang, P., Yang, H., Qiu, W., Wang, S., & Li, C. (2014). Optimal approach on net routing for VLSI physical design based on Tabu-ant colonies modeling. *Applied Soft Computing*, 21, 376–381
25. Ghosh, S., & Samanta, S. (2014). Fixed structure compensator design using a constrained hybrid evolutionary optimization approach. *ISA transactions*, 53(4), 1119–1130
26. Karimullah, S., & Vishnuvardhan, D. (2020). Iterative Analysis of Optimization Algorithms for Placement and Routing in ASIC Design. *ICDSMLA 2019* (pp. 1947–1958). Singapore: Springer
27. Kourany, T., Ghoneima, M., Hegazi, E., & Ismail, Y. (2017). Passiot: a pareto-optimal multi-objective optimization approach for synthesis of analog circuits using sobol' indices-based engine. *Integration*, 58, 9–21
28. Chen, J., Liu, Y., Zhu, Z., & Zhu, W. (2017). An adaptive hybrid memetic algorithm for thermal-aware non-slicing VLSI floorplanning. *Integration*, 58, 245–252
29. Sivaranjani, P., & Kumar, A. S. (2015). Thermal-aware non-slicing VLSI floorplanning using a smart decision-making PSO-GA based hybrid algorithm. *Circuits Systems and Signal Processing*, 34(11), 3521–3542
30. Karimullah, S., & Vishnuvardhan, D. (2020). *Experimental Analysis of Optimization Techniques for Placement and Routing in ASIC Design*. In *ICDSMLA 2019* (pp. 908–917). Singapore: Springer

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Shaik Karimullah Shaik Karimullah received his B.Tech Degree in ECE from MeRITS, Udagiri in 2009 and M.Tech Degree in VLSI System Design from Madina Engineering College, Kadapa in 2011. Currently he is a research scholar at JNTUA Ananthapuramu. He worked as Assistant Professor in Madina Engineering College from 2011–12. From 2012 to till now he is working as Assistant Professor in Electronics and Communication Engineering department in Annamacharya Institute of Technology and Sciences, Rajampet. During this period he published papers in peer reviewed journals and also published patents. His research areas include VLSI System Design, Analog Circuits, Wireless Communication. He is a Life member of Institute of Engineers (LM-IE), Kolkata, India. He is also a Life member of the Indian Society of Technical Education (LM-ISTE)



Dr.D.VishnuVardhan Dr D.VishnuVardhan received his B.Tech Degree in ECE from RGM College of Engineering in 2000 and M.Tech from JNTU Kakinada in 2005 and Ph. D degree in 2015 from JNTU Ananthapur. From June, 2001 to November, 2002 and January, 2004 to August, 2006 he worked as Lecturer at JNTU College of Engineering, Anantapur. From August, 2006 to November, 2006 he worked as Lecturer at JNTU College of Engineering, Pulivendula. During November, 2006 to 2020 he worked as Assistant Professor at JNTUA. Currently he is working as an Associate Professor and Head in the Department of ECE at JNTUA Ananthapuramu. During this period he held responsibilities such as Head of the Department, TEQIP-III Coordinator, Placement Officer. His research interest includes Embedded Systems, Image Processing and VLSI System Design. He is a Life member of Institute of Engineers (LM-IE), Kolkata, India. He is also a

Life member of the Indian Society of Technical Education (LM-ISTE) and also member in International Association of Engineers (IAENG).



Dr. Vidhyacharan Bhaskar Dr. Vidhyacharan Bhaskar received the B. Sc. degree in Mathematics from the University of Madras, Chennai, India in 1992, M.E. degree in Electrical & Communication Engineering from the Indian Institute of Science, Bangalore in 1997, and the M.S.E. and Ph.D. degrees in Electrical Engineering from the University of Alabama in Huntsville in 2001 and 2002, respectively. During 2002-2003, he was a Post Doctoral fellow with the Communications research group at the University of Toronto, Canada, where he worked on the applications of space-time coding for wireless communication systems. During 2003-2006, he was an Associate Professor in the Department of Information Systems and Telecommunications at the University of Technology of Troyes, France. From Jan. 2007 to May 2014, he was a Full Professor in the Department of Electronics and Communication Engineering at S.R.M. University, Kattankulathur, India. Since 2015, he is a Professor in the Department of Electrical and

Computer Engineering at San Francisco State University, San Francisco, California, USA. His research interests include MIMO wireless communications, signal processing, error control coding and queuing theory. He has published 133 Refereed Journal papers, presented around 72 Conference papers in various International Conferences over the past 20 years, published a book on "Adaptive Rate Coding for A-CDMA Systems" in Jan 2011, a book on "Higher-Order s-to-z mapping functions for digital filters," in March 2013, and has also co-authored a book on MATLAB in 1998. He has 930 Google scholar citations. He has advised 50 Masters students, 11 Doctoral students, and 3 Senior design projects. He is an IEEE Senior member (SM-IEEE) and is a member of IET (M-IET, UK). He is a Fellow of Institute of Electronics and Telecommunication Engineers (F-IETE), and a Fellow of Institute of Engineers (F-IE), Kolkata, India. He is also a Life member of the Indian Society of Technical Education (LM-ISTE) and a member of the Indian Science Congress (M-ISC).

Authors and Affiliations

Shaik Karimullah¹ · D. Vishnuvardhan² · Vidhyacharan Bhaskar³

✉ Vidhyacharan Bhaskar
vcharan@gmail.com

Shaik Karimullah
munnu483@gmail.com

D. Vishnuvardhan
vishnu.ece@jntua.ac.in

¹ Jawaharlal Nehru Technological University, Anantapur, A P, India

- ² Department of Electronics and Communication Engineering, Jawaharlal Nehru Technological University, Anantapur, A P, India
- ³ Dept. of Electrical and Computer Engineering, San Francisco State University, 94132 San Francisco, CA, USA